

10. Processor Management & CPU Scheduling

Computer Science Fundamentals | Module 3: Process Subsystem | CS_Fundamental-00

Interview Memory Hook: Program hard disk par rakhi ek band kitaab hai. Jab use khol kar execute karte hain, tab woh **Process** banti hai! OS ka Processor Manager decide karta hai ki kaun sa process kab tak CPU core par chalega.

10.1 Program vs Process & PCB

- **Program:** Passive code file on disk (app.exe).
- **Process:** Active entity in execution with Program Counter, Stack, Heap, Registers, and PCB.
- **PCB (Process Control Block):** Process ID, State, CPU Registers, Memory limits.

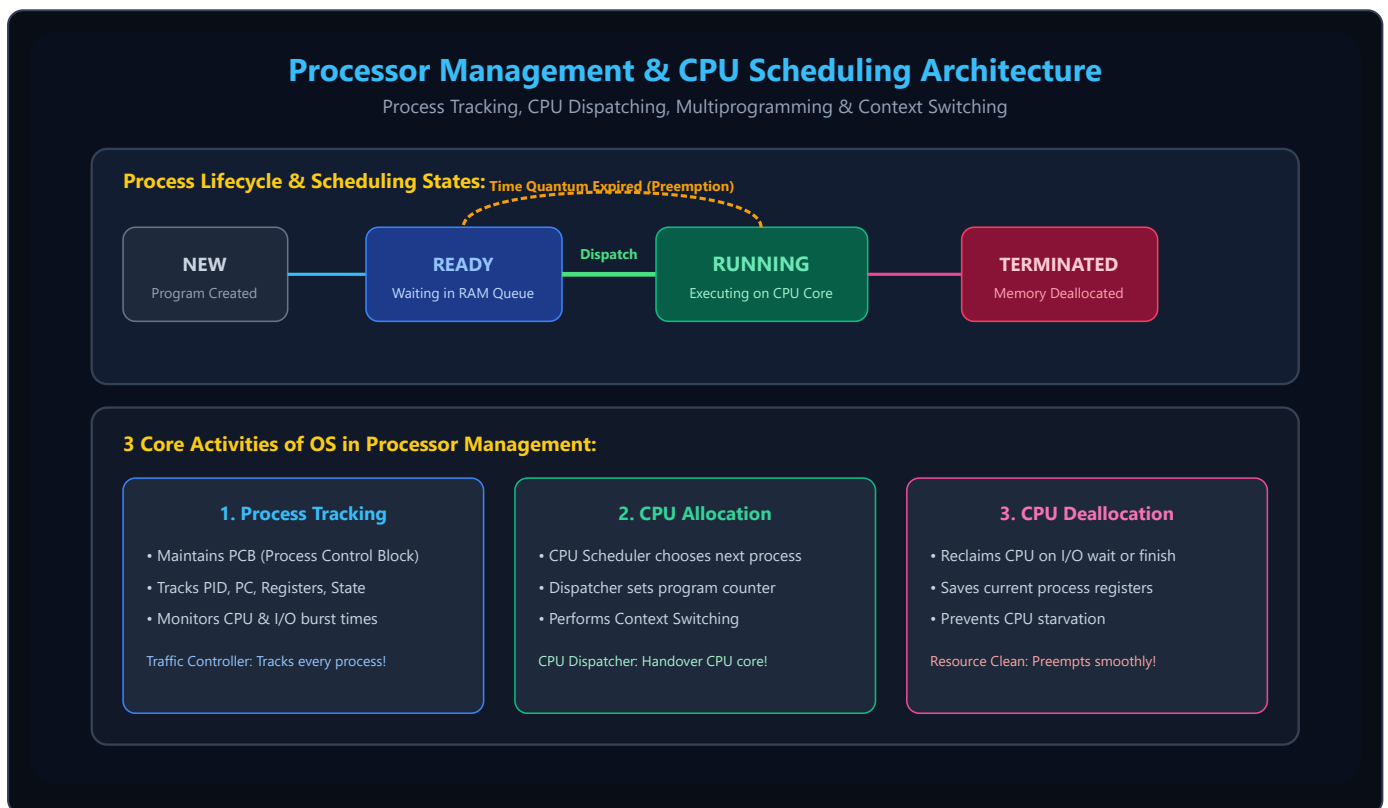


Fig 10.1: Process Lifecycle & CPU Scheduling Dispatcher

10.2 Multiprogramming & Scheduling

CPU utilization maximize karne ke liye jab chal raha process I/O wait karta hai, OS turant doosre ready process ko CPU dispatch kar deta hai.

10.3 OS ki 3 Core Activities

1. **Process Tracking:** PCB aur process states (New, Ready, Running, Waiting, Terminated) monitor karna.
2. **CPU Allocation:** Ready queue se process chun kar CPU core assign karna (Context Switch).
3. **CPU Deallocation:** Task finish hone par ya time quantum expire hone par CPU reclaim karna.

10.4 Numerical / Interview Thought Process: Context Switch Latency Trace

Context Switch Execution Steps (Process P1 → P2):

1. Timer interrupt trigger hota hai ($t = 0 \mu s$), CPU enters Kernel Mode.
2. Current process P1 ke registers aur program counter PCB1 mein save hote hain ($t = 1 \mu s$).
3. Scheduler agla process P2 choose karta hai ($t = 2 \mu s$).
4. PCB2 se saved state CPU registers mein reload hoti hai ($t = 3 \mu s$).
5. Mode bit 1 set karke P2 execute hota hai ($t = 4 \mu s$). Total overhead = $4 \mu s$ non-productive time!